

Module 5

Editors, Projects & Archives

Learning objectives

- Use a consistent src/tb/scripts/build layout
- Find files and search with grep -R
- Create and extract tar.gz archives
- Back up projects before major changes

Prerequisites

- Module 4 (basic scripts)

Learning path

Learning Path

Diagram: Learning Path

(render with mmdc when available)

flowchart LR

A[Editors] --> B[Project layout]

B --> C[find & grep]

C --> D[tar & zip]

D --> E[Backups]

E --> F[Module 6]

Course context

- RTL projects mirror course repo structure
- Exclude build/ from archives and Git
- `grep -R` finds signals across a tree

Core topics

Self-check

```
$ ./scripts/module5.sh --check
```

Terminal: module_check

```

[0;36m=====
[0;36mModule 5: Environment self-check
[0;36m=====

[0;32m[OK] tar available
[0;32m[OK] zip available
[0;32m[OK] unzip available
[0;32m[OK] find available
[0;32m[OK] grep available
[0;32m[OK] module5 directory exists: /home/yongfu/proj/universal-verification-methodology/
[0;32m[OK] project_structure/sample_project exists
[0;32m[OK] backup/backup_demo.sh exists

[0;32m[OK] All required checks passed.
```

Create archive

```
tar czf project_$(date +%Y%m%d).tar.gz project_dir/  
tar xzf project.tar.gz
```


Project structure

```
$ cd module5/examples/project_structure/sample_project && find . -type  
f | head -8
```

Terminal: project_tree

```
./build/.gitkeep  
./docs/notes.md  
./scripts/build.sh  
./src/main.v  
./tb/test_main.v
```

tar archives

```
$ cd module5/examples/tar_archives && tar czf /tmp/m5_demo.tar.gz  
demo_dir/ && tar tzf /tmp/m5_demo.tar.gz | head -5
```

Terminal: tar_list

```
demo_dir/  
demo_dir/file1.txt  
demo_dir/file2.txt  
demo_dir/subdir/  
demo_dir/subdir/file3.txt
```

Practice

Exercise scaffold

```
./scripts/module5.sh --scaffold
```

Exercises

- Scaffold `unix_course_project/`
- `grep -R` in `sample_project`
- Create a `tar.gz` backup

Summary & next steps

- Organize src/tb/scripts/build
- Archives for sharing
- Next: Module 6 — Git basics
- Next: Module 6: Git Basics for Course Work